

February 19, 2018

School District 68 (Nanaimo-Ladysmith)
395 Wakesiah Road
Nanaimo, BC V9R 3K6

ISSUED FOR USE
FILE: 704-ENW.VENW03150-01
Via Email: BHackwood@sd68.bc.ca

Attention: Brian Hackwood, Maintenance Manager

Subject: Domestic Water Testing (Lead) Inventory – Mountain View Elementary School

1.0 INTRODUCTION

Tetra Tech Canada Inc. (Tetra Tech) was retained School District 68 Nanaimo-Ladysmith (SD 68) to conduct a domestic water testing inventory at Mountain View Elementary School located in SD 68. Tetra Tech understands that the BC Ministry of Education has issued a directive to protect drinking water. The directive requires that a systematic investigation of public drinking water supplies at select schools in the District be undertaken. The investigation is based on procedures set forth by Health Canada modified per the Vancouver Island Health Authority (VIHA) guidelines, to ascertain risk and mitigation.

Tetra Tech understands that the sampling schedule, collection, testing and reporting of results needs to be completed by March 1, 2018 in order to allow sufficient time for SD 68 to implement mitigation measures prior to its deadline of March 31, 2018.

Mr. Brian Hackwood, Maintenance Manager with SD 68, provided Tetra Tech with authorization to proceed with the inventory on January 18, 2018.

2.0 METHODOLOGY

Tetra Tech completed the domestic water testing inventory program at Mountain View Elementary School on January 29th, 2018. The 2018 sampling program was conducted as per the protocols established during the 2016 program. The methodologies employed during the field program are detailed in the following subsections.

2.1 Sampling Locations

Tetra Tech reviewed plans for the facility prior to commencing the field work to identify potential sampling locations. The facility was then assessed in the field and sampling locations were selected based on the probability of human consumption at a location. The sampling locations included one point that was closest to the location where the water supply enters the building, one that is the furthest point from where the water supply enters the building and from points where human consumption of water occurred or was reasonably likely to occur. The sampling locations for Mountain View Elementary School are shown on the attached Figure 1.

Hallway drinking fountains and kitchen sinks were all considered to have a high probability of human consumption of water and were always sampled. Sinks with visible evidence of human consumption of water, such as water bottles, cups, or electric kettles were also considered to have a high probability of human consumption of water and were sampled. Although classroom sinks (including those with water fountains) were considered to have a moderate

to high probability of human consumption of water, only representative samples were collected as per the direction of SD 68 Maintenance Manager, Mr. Brian Hackwood. Classroom laboratory and art room sinks, where present, were considered to have a low probability of human consumption of water so only representative samples were collected. Finally, washrooms and utility sinks, unless there was other evidence of human consumption of water (such as an electric kettle) were considered to be a low probability of human consumption of water and only representative samples were collected.

2.2 Drinking Water Sampling

Sampling was conducted in the early hours of Monday, January 29th, 2018 in order collect water samples representative of an approximate worse-case scenario of water that had remained in contact with the school's plumbing over the course of a weekend. Two samples were collected at each sample location; the first collected immediately prior to any water line flushing (0 second sample); the second collected after thirty seconds of water line flushing (30 second sample).

The process for the sequence of analysis for a sample location is as follows:

- Only the pre-flush (0 second) sample is initially submitted for laboratory analysis;
- If the analytical result exceeds the *Guidelines for Canadian Drinking Water Quality* (GCDQG) Maximum Allowable Concentration (MAC), the 30 second sample would be submitted for further analysis; and
- If the 30 second sample analytical result exceeds the GCDQG MAC, additional samples would be collected after flushing with cold water for 2 minutes and 5 minutes at a subsequent sampling event and both samples submitted for analysis.

Water samples were collected directly from the sample point into clean, labeled, new laboratory-supplied. After collecting the water each sample then had preservative added for total lead. Sampling personnel wore new nitrile gloves prior to collecting each sample. Samples were kept in a cooler with ice after collection until being brought back to Tetra Tech's Nanaimo office, where samples not immediately submitted to the laboratory were stored in refrigerated conditions.

2.3 Analytical Testing

Samples were analyzed by Maxxam Analytics in Burnaby, British Columbia. Maxxam is a Canadian Association for Laboratory Accreditation (CALA) accredited laboratory that is qualified to analyze the samples using British Columbia Ministry of Environment (MOE)-approved procedures. All water samples submitted were analyzed for total lead.

2.4 Quality Assurance / Quality Control

During the sampling program, Tetra Tech implemented a Quality Assurance/Quality Control (QA/QC) program to ensure the integrity of the sampling methodology and analytical testing. The QA/QC program adhered to Tetra Tech's in-house Quality Management System (QMS), which was designed to generate representative samples, minimize the potential for cross-contamination between sampling locations and samples, and reduce the potential for systematic bias.

The QA/QC program included the following tasks:

- Recording the results of field activities in the field concurrently with the activities;

- Use of clean, new sampling gloves at each sampling location;
- Placing samples into new, labeled laboratory-supplied containers;
- Transporting samples to Maxxam in chilled coolers using chain-of-custody procedures;
- Using a Canadian Association for Laboratory Accreditation (CALA) accredited laboratory that is qualified to analyze the samples using MOE-approved procedures;
- Independently verifying the sample concentrations flagged by Maxxam as being greater than Health Canada guidelines by uploading laboratory results using ESdat, an environmental data management software, to minimize transcription errors; and
- Conducting a review of this report by a qualified senior Tetra Tech professional to ensure that the report meets Tetra Tech technical and reporting requirements.

Laboratory Quality Assurance / Quality Control Program

Water samples were submitted to Maxxam, a CALA accredited laboratory. Laboratory testing was conducted using methods outlined in the British Columbia Environmental Laboratory Manual. Laboratory QA/QC reports are attached to the laboratory reports presented in Appendix B. Samples included in the QA/QC reports consist of laboratory batches and will include random samples from the lab report and potentially other projects to complete a batch.

The Laboratories will note any sample deficiencies, such as unacceptable headspace, broken jars or bottles, etc. As well, the laboratory will measure the temperature of samples received by the laboratory in Burnaby.

Laboratory Duplicate – Relative Percent Difference (RPD)

A second aliquot is obtained from a randomly chosen sample. The aliquot is processed and the results expressed as the RPD between the two results. The purpose of the laboratory duplicate is to evaluate analytical precision and sample homogeneity. Tetra Tech formed the duplicate samples by alternately placing approximately 10% of the sample volume into the original sample container and then placing the same amount into the duplicate sample container. Tetra Tech continued placing additional aliquots of approximately 10% of the sample volume into each container until both containers were filled. RPDs should only be calculated and assessed when both the sample and the duplicate concentration is greater than five times the reportable detection limit (RDL), referred to as the Practical Quantification Limit (PQL).

The RPD calculations are discussed in Section 5.0. At Mountain View Elementary School, duplicate sample 18MV19-0s was collected at 18MV18-0s.

3.0 ASSESSMENT STANDARDS

As per the guidance from the Vancouver Island Health Authority (VIHA), Tetra Tech compared the sample analytical results to the *Guidelines for Canadian Drinking Water Quality* (GCDQG) published by Health Canada, February 2017. The guidelines list a Maximum Acceptable Concentration (MAC) for lead of 0.010 mg/L (10 µg/L). The MAC for lead is based on chronic effects and is intended to apply to average concentrations in water consumed for extended periods. No immediately toxic concentration for lead is listed, however exposure to lead should nevertheless be kept to a minimum.

4.0 ANALYTICAL RESULTS

Tetra Tech collected water samples from Mountain View Elementary School on January 29th, 2018. A total of 18 sample locations were identified; two samples were collected at each location (i.e., 0 second sample and 30 second sample). 18 pre-flush (0 second) samples were submitted for laboratory analysis of total lead.

Sampling locations are shown on Figure 1. Laboratory testing results for Mountain View Elementary School are summarized in the table below. The complete laboratory certificate is provided as Appendix B with Mountain View Elementary School results found on included on pages 4-5.

Table 1: Laboratory Testing Results

Sample ID	Sample Date	MAC	Total Lead (µg/L)
0 Second Samples			
18MV01-0s	1/29/2018	10 µg/L	0.49
18MV02-0s	1/29/2018		4.81
18MV03-0s	1/29/2018		0.72
18MV04-0s	1/29/2018		<0.20
18MV05-0s	1/29/2018		0.95
18MV06-0s	1/29/2018		2.70
18MV07-0s	1/29/2018		<0.20
18MV08-0s	1/29/2018		0.20
18MV09-0s	1/29/2018		<0.20
18MV10-0s	1/29/2018		0.59
18MV11-0s	1/29/2018		<0.20
18MV12-0s	1/29/2018		0.70
18MV13-0s	1/29/2018		1.29
18MV14-0s	1/29/2018		0.21
18MV15-0s	1/29/2018		0.21
18MV16-0s	1/29/2018		<0.20
18MV17-0s	1/29/2018		0.59
18MV18-0s	1/29/2018		0.28
18MV19-0s	1/29/2018		<0.20
*duplicate sample 18MV19-0s was collected at 18MV18-0s			

All eighteen pre-flush (0 second) samples and duplicate contained concentrations of total lead less than the GCDQG MAC of 10 µg/L (0.010 mg/L).

5.0 DISCUSSION AND RECOMMENDATIONS

Tetra Tech's sampling program was based upon guidance from the Ministry of Health, found in the document *Guidance on Controlling Corrosion in Drinking Water Distribution Systems* (2009). The rationale is that for each sampling point, if the pre-flush (0 second) sample (Tier 1) contained elevated lead concentrations, it could indicate that the faucet or fittings are the likely be the source of lead. If a subsequent 30 second flush sample (Tier 2) contained elevated lead concentrations, the source of the lead would likely be the piping (plumbing) leading to the faucet; whereas low lead concentrations in the 30 second sample would further indicate that the source was likely the faucet and fittings. Finally, a 5 minute flush sample (if required) should be drawing water directly from the water

supply piping within the building and would indicate if flushing is feasible for lowering the lead concentration in water within the building.

The Health Canada guidance recommend that Tier 2 sampling (30 second samples) take place when Tier 1 sampling identifies more than 10% of sites with lead concentrations above the MAC, and then only at the 10% of sampling sites with the highest lead concentration. Rather, Tetra Tech ran every 30 second sample for locations where the 0 second sample was above the MAC to show that flushing was adequate to lower the lead concentration in the drinking water at each point of concern.

The guidance from the Ministry of Health recommended that samples be collected after the sampling points had been stagnant for a minimum of 8 hours but not longer than 24 hours in order to simulate the worst case daily scenario for lead in drinking water consumption. Based on guidance from VIHA, SD 68 directed Tetra Tech to collect samples Monday mornings prior to any staff or students arriving at the facilities in order to simulate a worst-case scenario for stagnant water. As such, lead concentrations reported represent what could be expected following a weekend and would likely be lower on subsequent weekday mornings.

All pre-flush (0 second) samples collected at Mountain View Elementary School contained concentrations of lead below the GCDQG MAC indicating that lead sources in the supply water to this facility is not a concern.

During sample collection, Tetra Tech noted signage throughout the facility stating “Water Quality – First thing in the morning... Run the water for two minutes before drinking. Throughout the day... Let the water run until it is cold before drinking.” Despite the fact the analytical results indicate that all possible drinking locations at Mountain View Elementary School had low lead concentrations, Tetra Tech recommends that this signage be maintained at each point where drinking water could be consumed and that this procedure continues to be followed as it promotes drinking water safety awareness.

At Mountain View Elementary School duplicate sample 18MV19-0s was collected at 18MV18-0s. The total lead concentration for both samples was below 5 times RDL of 0.20 µg/L, and as such the RPD was not calculated. Given, Maxxam's internal QA/QC process, and that the remaining duplicate samples collected throughout the program have generally met the 30% screening threshold recommended by BC Ministry of Environment Q&A, and BC Environmental Laboratory Manual, Tetra Tech considers the analytical results to be valid and re-sampling not necessary.

6.0 SUMMARY AND CONCLUSIONS

All pre-flush (0 second) samples collected at Mountain View Elementary School contained concentrations of total lead below the GCDQG MAC of 10µg/L (0.010 mg/L).

Tetra Tech recommends that SD 68 continue with its ongoing procedure of conducting a 2 minute flush at each drinking water consumption point each morning; and running taps/faucets until cold prior to consuming water. Tetra Tech noted signage at most drinking water consumption points stating “Water Quality – First thing in the morning... Run the water for two minutes before drinking. Throughout the day... Let the water run until it is cold before drinking.”

Tetra Tech recommends that the facility be inspected on a routine basis to ensure that the above noted signage is present and in good condition at each point where drinking water could be consumed. Tetra Tech further recommends that a bulletin be provided to staff summarizing the drinking water quality results at the facility and reminding them of the above procedure. Staff should then instruct students and visitors in the drinking water procedure.

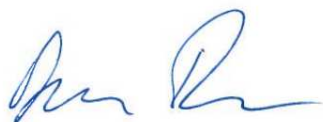
7.0 CLOSURE

This report has been prepared based on the scope of work and for the use of School District 68, which includes distribution as required for the purposes for which this assessment was commissioned. The assessment has been carried out in accordance with generally accepted professional practice. No other warranty is made, either express or implied. Professional judgment has been applied in developing the recommendations in this report.

This report was prepared by personnel with professional experience in investigations of this nature and who specifically conducted the investigations at this Site. Reference should be made to the 'Geoenvironmental Report – Limitations on the Use of this Report' attached in Appendix A that forms a part of this report.

We trust this report meets your present requirements. If you have any questions or comments, please contact the undersigned.

Respectfully submitted,
Tetra Tech Canada Inc.



Darren Thomas, B.A.Sc., EIT.
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Environment Practice
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Darren.Thomas@tetrattech.com



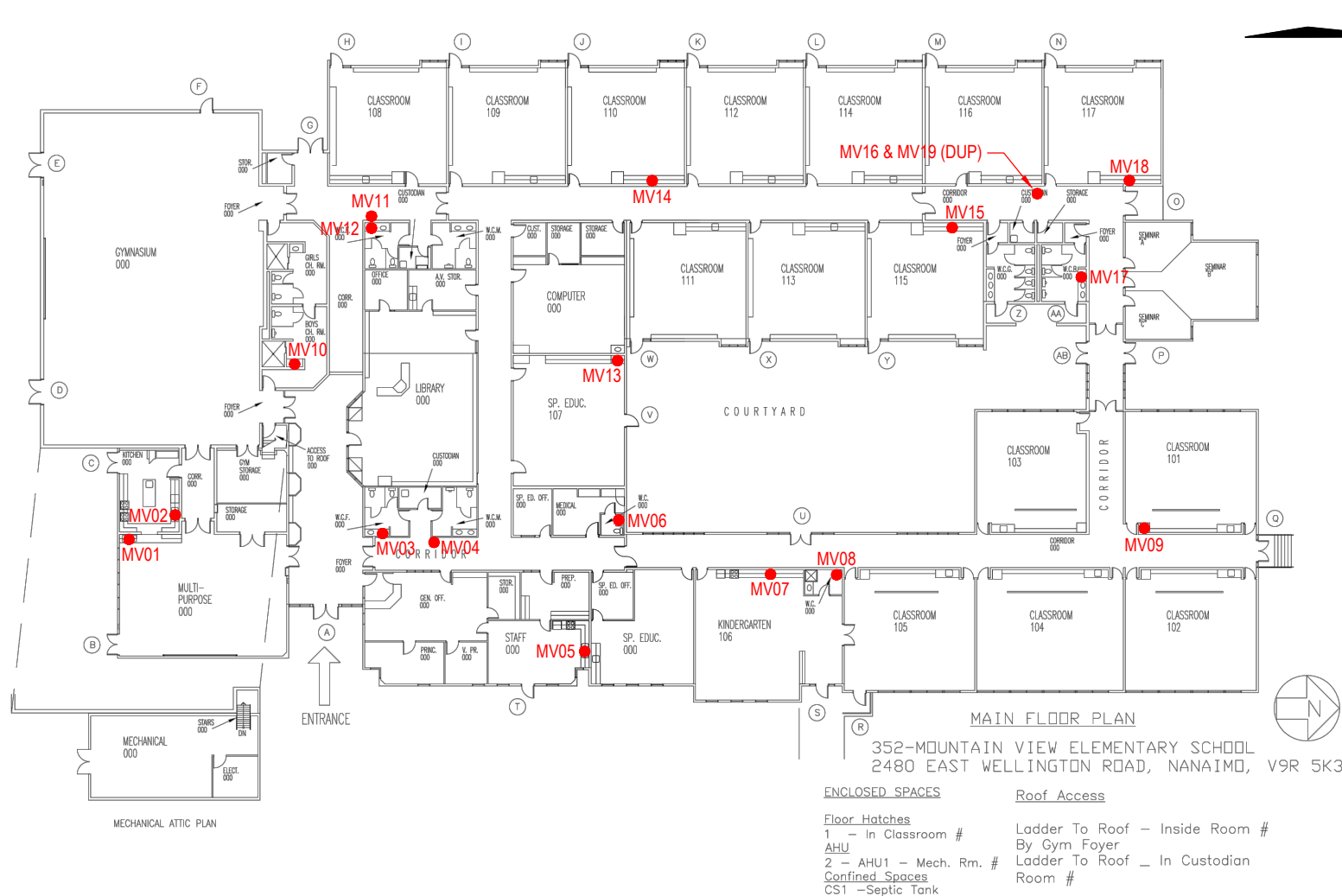
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/dr

Attachments: Figure 1 - Mountain View Elementary School Sample Locations
Appendix A - Limitations on the Use of this Document
Appendix B - Laboratory Report

FIGURES

Figure 1 Mountain View Elementary School Sample Locations



S:\ACAD\MASTER CAD FILES 2013\2013 CAD\2013 Floor Plans\Elementary Schools\352 Mountain View\Floor Plan\352 Mt. View 2013.dwg

LEGEND: - SAMPLE LOCATION	NOTES BASED ON DRAWING PROVIDED BY CLIENT DRAWING IS NOT TO SCALE STATUS ISSUED FOR USE	CLIENT SCHOOL DISTRICT NO 68	DOMESTIC WATER TESTING LEAD INVENTORY				
			352 - MOUNTAIN VIEW ELEMENTARY SCHOOL 2480 EAST WELLINGTON ROAD, NANAIMO BC				
		TETRA TECH	PROJECT NO. ENW.VENW03150-01	DWN MRV	CKD DT	REV 0	Figure 1
	OFFICE EDM	DATE February 1, 2018					

APPENDIX A

LIMITATIONS ON THE USE OF THIS DOCUMENT

LIMITATIONS ON USE OF THIS DOCUMENT

GEOENVIRONMENTAL

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Both electronic file and/or hard copy versions of TETRA TECH's Instruments of Professional Service shall not, under any circumstances, be altered by any party except TETRA TECH. TETRA TECH's Instruments of Professional Service will be used only and exactly as submitted by TETRA TECH.

Electronic files submitted by TETRA TECH have been prepared and submitted using specific software and hardware systems. TETRA TECH makes no representation about the compatibility of these files with the Client's current or future software and hardware systems.

1.3 STANDARD OF CARE

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consistent with the level of skill ordinarily exercised by members of the profession currently practicing under similar conditions in the jurisdiction in which the services are provided. Professional judgment has been applied in developing the conclusions and/or recommendations provided in this Professional Document. No warranty or guarantee, express or implied, is made concerning the test results, comments, recommendations, or any other portion of the Professional Document.

If any error or omission is detected by the Client or an Authorized Party, the error or omission must be immediately brought to the attention of TETRA TECH.

1.4 DISCLOSURE OF INFORMATION BY CLIENT

The Client acknowledges that it has fully cooperated with TETRA TECH with respect to the provision of all available information on the past, present, and proposed conditions on the site, including historical information respecting the use of the site. The Client further acknowledges that in order for TETRA TECH to properly provide the services contracted for in the Contract, TETRA TECH has relied upon the Client with respect to both the full disclosure and accuracy of any such information.

1.5 INFORMATION PROVIDED TO TETRA TECH BY OTHERS

During the performance of the work and the preparation of this Professional Document, TETRA TECH may have relied on information provided by persons other than the Client.

While TETRA TECH endeavours to verify the accuracy of such information, TETRA TECH accepts no responsibility for the accuracy or the reliability of such information even where inaccurate or unreliable information impacts any recommendations, design or other deliverables and causes the Client or an Authorized Party loss or damage.

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This Professional Document is based solely on the conditions presented and the data available to TETRA TECH at the time the data were collected in the field or gathered from available databases.

The Client, and any Authorized Party, acknowledges that the Professional Document is based on limited data and that the conclusions, opinions, and recommendations contained in the Professional Document are the result of the application of professional judgment to such limited data.

The Professional Document is not applicable to any other sites, nor should it be relied upon for types of development other than those to which it refers. Any variation from the site conditions present, or variation in assumed conditions which might form the basis of design or recommendations as outlined in this report, at or on the development proposed as of the date of the Professional Document requires a supplementary investigation and assessment.

TETRA TECH is neither qualified to, nor is it making, any recommendations with respect to the purchase, sale, investment or development of the property, the decisions on which are the sole responsibility of the Client.

1.7 NOTIFICATION OF AUTHORITIES

In certain instances, the discovery of hazardous substances or conditions and materials may require that regulatory agencies and other persons be informed and the client agrees that notification to such bodies or persons as required may be done by TETRA TECH in its reasonably exercised discretion.

APPENDIX B

LABORATORY REPORT

Your Project #: ENW.VENW03150

Attention: Ben Barton

TETRA TECH CANADA INC.
#1 - 4376 BOBAN DRIVE
NANAIMO, BC
Canada V9T 6A7

Your C.O.C. #: 545893-24-01, 545893-03-01, 545893-04-01, 545893-01-01, 545893-02-01

Report Date: 2018/02/06

Report #: R2510579

Version: 1 - Final

CERTIFICATE OF ANALYSIS

MAXXAM JOB #: B807272

Received: 2018/01/30, 08:48

Sample Matrix: DRINKING WATER
Samples Received: 48

Analyses	Quantity	Date Extracted	Date Analyzed	Laboratory Method	Analytical Method
Elements by CRC ICPMS (total)	48	N/A	2018/02/01	BBY7SOP-00003,	EPA 6020b R2 m

Remarks:

Maxxam Analytics' laboratories are accredited to ISO/IEC 17025:2005 for specific parameters on scopes of accreditation. Unless otherwise noted, procedures used by Maxxam are based upon recognized Provincial, Federal or US method compendia such as CCME, MDDELCC, EPA, APHA.

All work recorded herein has been done in accordance with procedures and practices ordinarily exercised by professionals in Maxxam's profession using accepted testing methodologies, quality assurance and quality control procedures (except where otherwise agreed by the client and Maxxam in writing). All data is in statistical control and has met quality control and method performance criteria unless otherwise noted. All method blanks are reported; unless indicated otherwise, associated sample data are not blank corrected.

Maxxam Analytics' liability is limited to the actual cost of the requested analyses, unless otherwise agreed in writing. There is no other warranty expressed or implied. Maxxam has been retained to provide analysis of samples provided by the Client using the testing methodology referenced in this report. Interpretation and use of test results are the sole responsibility of the Client and are not within the scope of services provided by Maxxam, unless otherwise agreed in writing.

Solid sample results, except biota, are based on dry weight unless otherwise indicated. Organic analyses are not recovery corrected except for isotope dilution methods.

Results relate to samples tested.

This Certificate shall not be reproduced except in full, without the written approval of the laboratory.

Reference Method suffix "m" indicates test methods incorporate validated modifications from specific reference methods to improve performance.

* RPDs calculated using raw data. The rounding of final results may result in the apparent difference.

Your Project #: ENW.VENW03150

Attention: Ben Barton

TETRA TECH CANADA INC.
#1 - 4376 BOBAN DRIVE
NANAIMO, BC
Canada V9T 6A7

Your C.O.C. #: 545893-24-01, 545893-03-01, 545893-04-01, 545893-01-01, 545893-02-01

Report Date: 2018/02/06
Report #: R2510579
Version: 1 - Final

CERTIFICATE OF ANALYSIS

MAXXAM JOB #: B807272

Received: 2018/01/30, 08:48

Encryption Key

Please direct all questions regarding this Certificate of Analysis to your Project Manager.

Letitia Prefontaine, B.Sc., Senior Project Manager

Email: LPrefontaine@maxxam.ca

Phone# (604)639-2616

=====

Maxxam has procedures in place to guard against improper use of the electronic signature and have the required "signatories", as per section 5.10.2 of ISO/IEC 17025:2005(E), signing the reports. For Service Group specific validation please refer to the Validation Signature Page.

Maxxam Job #: B807272
Report Date: 2018/02/06

TETRA TECH CANADA INC.
Client Project #: ENW.VENW03150

ELEMENTS BY ATOMIC SPECTROSCOPY (DRINKING WATER)

Maxxam ID			SW7332	SW7333	SW7334	SW7335	SW7336	SW7337		
Sampling Date			2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29		
COC Number			545893-24-01	545893-24-01	545893-24-01	545893-24-01	545893-24-01	545893-24-01		
	UNITS	MAC	18 ST1-OS	18 ST02-OS	18 ST03-OS	18 ST04-OS	18 ST05-OS	18 ST06-OS	RDL	QC Batch

Total Metals by ICPMS

Total Lead (Pb)	ug/L	10	8.86	8.50	0.52	0.70	10.5	0.58	0.20	8898699
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No Fill	No Exceedance
Grey	Exceeds 1 criteria policy/level
Black	Exceeds both criteria/levels

RDL = Reportable Detection Limit

Maxxam ID			SW7338	SW7339	SW7340	SW7353	SW7354	SW7355		
Sampling Date			2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29		
COC Number			545893-24-01	545893-24-01	545893-24-01	545893-03-01	545893-03-01	545893-03-01		
	UNITS	MAC	18 ST07-OS	18 ST08-OS	18 ST09-OS	18 DC01-OS	18 DC02-OS	18 DC03-OS	RDL	QC Batch

Total Metals by ICPMS

Total Lead (Pb)	ug/L	10	3.37	11.4	6.05	2.39	3.87	15.2	0.20	8898699
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No Fill	No Exceedance
Grey	Exceeds 1 criteria policy/level
Black	Exceeds both criteria/levels

RDL = Reportable Detection Limit

Maxxam ID			SW7356	SW7357	SW7358	SW7359	SW7360	SW7361		
Sampling Date			2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29		
COC Number			545893-03-01	545893-03-01	545893-03-01	545893-03-01	545893-03-01	545893-03-01		
	UNITS	MAC	18DC04-OS	18DC05-OS	18DC06-OS	18DC07-OS	18DC08-OS	18DC09-OS	RDL	QC Batch

Total Metals by ICPMS

Total Lead (Pb)	ug/L	10	12.2	5.00	8.40	5.57	21.0	68.6	0.20	8898701
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No Fill	No Exceedance
Grey	Exceeds 1 criteria policy/level
Black	Exceeds both criteria/levels

RDL = Reportable Detection Limit

Maxxam ID			SW7362	SW7363	SW7364	SW7365	SW7366	SW7367		
Sampling Date			2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29		
COC Number			545893-03-01	545893-04-01	545893-04-01	545893-04-01	545893-04-01	545893-04-01		
	UNITS	MAC	18DC10-OS	18DC11-OS	18DC12-OS	18DC13-OS	18DC14-OS	18DC15-OS	RDL	QC Batch

Total Metals by ICPMS

Total Lead (Pb)	ug/L	10	0.52	4.00	2.32	66.0	7.26	8.00	0.20	8898701
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No Fill	No Exceedance
Grey	Exceeds 1 criteria policy/level
Black	Exceeds both criteria/levels

RDL = Reportable Detection Limit

Maxxam Job #: B807272
Report Date: 2018/02/06

TETRA TECH CANADA INC.
Client Project #: ENW.VENW03150

ELEMENTS BY ATOMIC SPECTROSCOPY (DRINKING WATER)

Maxxam ID			SW7368	SW7369	SW7370	SW7371	SW7372	SW7373		
Sampling Date			2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29		
COC Number			545893-04-01	545893-04-01	545893-04-01	545893-04-01	545893-04-01	545893-01-01		
	UNITS	MAC	18DC16-OS	18DC17-OS	18DC18-OS	18DC19-OS	18DC20-OS	18MV01-OS	RDL	QC Batch

Total Metals by ICPMS

Total Lead (Pb)	ug/L	10	3.03	66.4	46.8	8.56	649	0.49	0.20	8898701
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No Fill	No Exceedance
Grey	Exceeds 1 criteria policy/level
Black	Exceeds both criteria/levels

RDL = Reportable Detection Limit

Maxxam Job #: B807272
Report Date: 2018/02/06

TETRA TECH CANADA INC.
Client Project #: ENW.VENW03150

ELEMENTS BY ATOMIC SPECTROSCOPY (DRINKING WATER)

Maxxam ID			SW7374		SW7375	SW7376	SW7377	SW7378		
Sampling Date			2018/01/29		2018/01/29	2018/01/29	2018/01/29	2018/01/29		
COC Number			545893-01-01		545893-01-01	545893-01-01	545893-01-01	545893-01-01		
	UNITS	MAC	18MV02-OS	QC Batch	18MV03-OS	18MV04-OS	18MV05-OS	18MV06-OS	RDL	QC Batch

Total Metals by ICPMS										
Total Lead (Pb)	ug/L	10	4.81	8898701	0.72	<0.20	0.95	2.70	0.20	8898711
No Fill	No Exceedance									
Grey	Exceeds 1 criteria policy/level									
Black	Exceeds both criteria/levels									
RDL = Reportable Detection Limit										

Maxxam ID			SW7379	SW7380	SW7381	SW7382	SW7383	SW7384		
Sampling Date			2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29		
COC Number			545893-01-01	545893-01-01	545893-01-01	545893-01-01	545893-02-01	545893-02-01		
	UNITS	MAC	18MV07-OS	18MV08-OS	18MV09-OS	18MV10-OS	18MV11-OS	18MV12-OS	RDL	QC Batch

Total Metals by ICPMS										
Total Lead (Pb)	ug/L	10	<0.20	0.20	<0.20	0.59	<0.20	0.70	0.20	8898711
No Fill	No Exceedance									
Grey	Exceeds 1 criteria policy/level									
Black	Exceeds both criteria/levels									
RDL = Reportable Detection Limit										

Maxxam ID			SW7385	SW7386	SW7387	SW7388	SW7389	SW7390		
Sampling Date			2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29	2018/01/29		
COC Number			545893-02-01	545893-02-01	545893-02-01	545893-02-01	545893-02-01	545893-02-01		
	UNITS	MAC	18MV13-OS	18MV14-OS	18MV15-OS	18MV16-OS	18MV17-OS	18MV18-OS	RDL	QC Batch

Total Metals by ICPMS										
Total Lead (Pb)	ug/L	10	1.29	0.21	0.21	<0.20	0.59	0.28	0.20	8898711
No Fill	No Exceedance									
Grey	Exceeds 1 criteria policy/level									
Black	Exceeds both criteria/levels									
RDL = Reportable Detection Limit										

Maxxam ID			SW7391		
Sampling Date			2018/01/29		
COC Number			545893-02-01		
	UNITS	MAC	18MV19-OS	RDL	QC Batch
Total Metals by ICPMS					
Total Lead (Pb)	ug/L	10	<0.20	0.20	8898711
No Fill	No Exceedance				
Grey	Exceeds 1 criteria policy/level				
Black	Exceeds both criteria/levels				
RDL = Reportable Detection Limit					

Maxxam Job #: B807272
Report Date: 2018/02/06

TETRA TECH CANADA INC.
Client Project #: ENW.VENW03150

GENERAL COMMENTS

Each temperature is the average of up to three cooler temperatures taken at receipt

Package 1	5.3°C
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Samples received with incomplete Chain of Custody. Sampling times not provided.

Chain of Custodies 545893-03-01, 545893-04-01, 545893-01-01 and 545893-02-01 not completed with signature/date in the "Relinquished by" line.

MAC: The guidelines that have been included in this report have been taken from the Canadian Drinking Water Quality Summary Table, February 2017.

Criteria A = Maximum Acceptable Concentration (MAC) / Criteria B = Aesthetic Objectives (AO) / Criteria C = Operational Guidance Values (OG)

It is recommended to consult these guidelines when interpreting your data since there are non-numerical guidelines that are not included on this report.

Turbidity Guidelines:

1. Chemically assisted filtration: less than or equal to 0.3 NTU in 95% of the measurements or 95% of the time each month. Shall not exceed 1.0 NTU at any time.
2. Slow sand / diatomaceous earth filtration: less than or equal to 1.0 NTU in 95% of the measurements or 95% of the time each month. Shall not exceed 3.0 NTU at any time.
3. Membrane filtration: less than or equal to 0.1 NTU in 99% of the measurements made or at least 99% of the time each calendar month. Shall not exceed 0.3 NTU at any time.

Results relate only to the items tested.

Maxxam Job #: B807272
Report Date: 2018/02/06

QUALITY ASSURANCE REPORT

TETRA TECH CANADA INC.
Client Project #: ENW.VENW03150

QC Batch	Parameter	Date	Matrix Spike		Spiked Blank		Method Blank		RPD	
			% Recovery	QC Limits	% Recovery	QC Limits	Value	UNITS	Value (%)	QC Limits
8898699	Total Lead (Pb)	2018/02/01	99	80 - 120	103	80 - 120	<0.20	ug/L	NC	20
8898701	Total Lead (Pb)	2018/02/01	101	80 - 120	99	80 - 120	<0.20	ug/L	0.89	20
8898711	Total Lead (Pb)	2018/02/01	96	80 - 120	99	80 - 120	<0.20	ug/L	3.5	20

Duplicate: Paired analysis of a separate portion of the same sample. Used to evaluate the variance in the measurement.

Matrix Spike: A sample to which a known amount of the analyte of interest has been added. Used to evaluate sample matrix interference.

Spiked Blank: A blank matrix sample to which a known amount of the analyte, usually from a second source, has been added. Used to evaluate method accuracy.

Method Blank: A blank matrix containing all reagents used in the analytical procedure. Used to identify laboratory contamination.

NC (Duplicate RPD): The duplicate RPD was not calculated. The concentration in the sample and/or duplicate was too low to permit a reliable RPD calculation (absolute difference $\leq 2 \times \text{RDL}$).

Maxxam Job #: B807272
Report Date: 2018/02/06

TETRA TECH CANADA INC.
Client Project #: ENW.VENW03150

VALIDATION SIGNATURE PAGE

The analytical data and all QC contained in this report were reviewed and validated by the following individual(s).



Andy Lu, Ph.D., P.Chem., Scientific Specialist

Maxxam has procedures in place to guard against improper use of the electronic signature and have the required "signatories", as per section 5.10.2 of ISO/IEC 17025:2005(E), signing the reports. For Service Group specific validation please refer to the Validation Signature Page.

Chain Of Custody Record

Page 1 of 5

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Maxxam Analytics International Corporation o/a Maxxam Analytics
4605 Canada Way, Burnaby, British Columbia Canada V5G 1K5 Tel: (604) 734 7276 Toll-free: 800-563-6266 Fax: (604) 731 2385 www.maxxam.ca

Chain Of Custody Record

25

INVOICE TO:		Report Information		Project Information	
Company Name	#1433 TETRA TECH CANADA INC.	Company Name		Quotation #	B60578
Contact Name	Ben Barton	Contact Name		P.O. #	
Address	#1 - 4376 BOBAN DRIVE NANAIMO BC V9T 6A7	Address		Project #	ENW.VENW03150
Phone	(250) 756-2256 x	Phone		Project Name	
Fax	(250) 756-2686 x	Fax		Site #	
Email	bbarton@eba.ca; EBA.Labdata@tetratech.com	Email		Sampled By	



B807272_COC

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Regulatory Criteria		Special Instructions		ANALYSIS REQUESTED (PLEASE BE SPECIFIC)												Turnaround Time (TAT) Required:			
☐ CSR ☐ CCME ☐ BC Water Quality ☐ Other																Please provide advance notice for rush projects			
SAMPLES MUST BE KEPT COOL (< 10°C) FROM TIME OF SAMPLING UNTIL DELIVERY TO MAXXAM														Regular (Standard) TAT: (will be applied if Rush TAT is not specified): Standard TAT = 5-7 Working days for most tests. Please note: Standard TAT for certain tests such as BOD and Dioxins/Furans are > 5 days - contact your Project Manager for details.					
														Job Specific Rush TAT (if applies to entire submission) 1 DAY ☐ 2 Day ☐ 3 Day ☐ Date Required: ☐ Rush Confirmation Number: ☐ (call lab for #)					
														# of Bottles		Comments			
Sample Barcode Label	Sample (Location) Identification	Date Sampled	Time Sampled	Matrix	Metals Field Filtered ? (Y/N)	Lead - Drinking Water													
1	18DC01-05	29-Jan-18		Water	✓														
2	18DC02-05				✓														
3	18DC03-05				✓														
4	18DC04-05				✓														
5	18DC05-05				✓														
6	18DC06-05				✓														
7	18DC07-05				✓														
8	18DC08-05				✓														
9	18DC09-05				✓														
10	18DC10-05				✓														
* RELINQUISHED BY: (Signature/Print)		Date: (YY/MM/DD)	Time	RECEIVED BY: (Signature/Print)		Date: (YY/MM/DD)	Time	# jars used and not submitted		Lab Use Only									
				POMMEL GORX		2018/01/30	08:46			Time Sensitive ☐ Temperature (°C) on Receipt 5.5.6 Custody Seal Intact on Cooler? ☐ Yes ☒ No									
* UNLESS OTHERWISE AGREED TO IN WRITING, WORK SUBMITTED ON THIS CHAIN OF CUSTODY IS SUBJECT TO MAXXAM'S STANDARD TERMS AND CONDITIONS. SIGNING OF THIS CHAIN OF CUSTODY DOCUMENT IS ACKNOWLEDGMENT AND ACCEPTANCE OF OUR TERMS WHICH ARE AVAILABLE FOR VIEWING AT WWW.MAXXAM.CA/TERMS.																			
* IT IS THE RESPONSIBILITY OF THE RELINQUISHER TO ENSURE THE ACCURACY OF THE CHAIN OF CUSTODY RECORD. AN INCOMPLETE CHAIN OF CUSTODY MAY RESULT IN ANALYTICAL TAT DELAYS.																			

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Page 4 of 5

B807272 COCRegular (Standard) TAT:

(will be applied if Rush TAT is not specified)

Standard TAT = 5-7 Working days for most tests.

Please note: Standard TAT for certain tests such as BOD and Dioxins/Furans are > 5 days - contact your Project Manager for details.

Job Specific Rush TAT (if applies to entire submission)

1 DAY ☐ 2 Day ☐ 3 Day ☐ Date Required _____

Rush Confirmation Number:

(call lab for #)

SAMPLES MUST BE KEPT COOL (< 10°C) FROM TIME OF SAMPLING UNTIL DELIVERY TO MAXXAM

Metals Field Filtered ? (Y / N)

Lead - Drinking Water

RECEIVED BY: (Signature/Print) _____

Date: (YYMMDD)

Time

# jars used and not submitted	
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Time Sensitive

Temperature: 2024-05-20 10:00

Lab Use Only

Custody Seal Intact on Cooler?

☐ Yes ☒ No

Yellow Client

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Chain Of Custody Record

5 of 5

INVOICE TO:		Report Information		Project Information	
Company Name	#1433 TETRA TECH CANADA INC.	Company Name		Quotation #	B60578
Contact Name	Ben Barton	Contact Name		P.O. #	
Address	#1 - 4376 BOBAN DRIVE	Address		Project #	ENW.VENW03150
	NANAIMO BC V9T 6A7			Project Name	
Phone	(250) 756-2256 x	Phone		Site #	
	Fax: (250) 756-2686 x	Fax		Sampled By	
Email	bbarton@eba.ca; EBA.Labdata@tetratech.com	Email			



B807272_COC

der #:

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Regulatory Criteria:		Special Instructions		ANALYSIS REQUESTED (PLEASE BE SPECIFIC)										Turnaround Time (TAT) Required:					
☐ CSR ☐ CCME ☐ BC Water Quality ☐ Other _____														Please provide advance notice for rush projects					
SAMPLES MUST BE KEPT COOL (< 10°C) FROM TIME OF SAMPLING UNTIL DELIVERY TO MAXXAM														Regular (Standard) TAT: (will be applied if Rush TAT is not specified): Standard TAT = 5-7 Working days for most tests. Please note: Standard TAT for certain tests such as BOD and Dioxins/Furans are > 5 days - contact your Project Manager for details.					
														Job Specific Rush TAT (if applies to entire submission) 1 DAY ☐ 2 Day ☐ 3 Day ☐ Date Required: _____ Rush Confirmation Number: _____ (call lab for #)					
														# of Bottles					
														Comments					
Sample Barcode Label	Sample (Location) Identification	Date Sampled	Time Sampled	Matrix	Metals Field Filtered ? (Y/N)	Lead - Drinking Water													
1	18mV11-05		Jan 29/18	Water	✓	✓													
2	18mV12-05				✓	✓													
3	18mV13-05				✓	✓													
4	18mV14-05				✓	✓													
5	18mV15-05				✓	✓													
6	18mV16-05				✓	✓													
7	18mV17-05				✓	✓													
8	18mV18-05				✓	✓													
9	18mV19-05				✓	✓													
10																			
* RELINQUISHED BY: (Signature/Print)		Date: (YY/MM/DD)	Time	RECEIVED BY: (Signature/Print)		Date: (YY/MM/DD)	Time	# Jars used and not submitted	Lab Use Only										
				[Signature]		29/1/18	08:48		Time Sensitive ☐	Temperature (°C) on Receipt 5.5.6	Custody Seal Intact on Cooler? ☐ Yes ☒ No								
* UNLESS OTHERWISE AGREED TO IN WRITING, WORK SUBMITTED ON THIS CHAIN OF CUSTODY IS SUBJECT TO MAXXAM'S STANDARD TERMS AND CONDITIONS. SIGNING OF THIS CHAIN OF CUSTODY DOCUMENT IS ACKNOWLEDGMENT AND ACCEPTANCE OF OUR TERMS WHICH ARE AVAILABLE FOR VIEWING AT WWW.MAXXAM.CA/TERMS.																		White: Maxxam Yellow: Client	
* IT IS THE RESPONSIBILITY OF THE RELINQUISHER TO ENSURE THE ACCURACY OF THE CHAIN OF CUSTODY RECORD. AN INCOMPLETE CHAIN OF CUSTODY MAY RESULT IN ANALYTICAL TAT DELAYS.																			

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